Exp: 11 **Routing at the Network Layer**

Aim:

To simulate static routing and RIP routing protocol configuration using Cisco Packet Tracer.

Requirements:

* Cisco Packet Tracer
* 2 routers, 2 switches, 4 PCs

Topology and Addressing:

* LAN 1: 192.168.1.0/24 (R1 Fa0/0 = 192.168.1.1)
* Link R1-R2: 10.0.0.0/30 (R1 Se0/0/0 = 10.0.0.1, R2 Se0/0/0 = 10.0.0.2)
* LAN 2: 192.168.2.0/24 (R2 Fa0/0 = 192.168.2.1)

(a) Static Routing:

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
R1# show ip route
```

Ping from a PC in LAN 1 to a PC in LAN 2 and verify the reply.

(b) RIP:

```
R1(config)# ~~roter~~ router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0
R1(config-router)# no auto-summary
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.0.0.0
R2(config-router)# no auto-summary
R1# show ip route          (routes marked R are learned through RIP)
```

Remove the static routes first (no ip route ...) so that RIP routes are used, then ping across the LANs.

Result:

Static routing and RIP were configured in Cisco Packet Tracer and end-to-end ~~connectivity~~ connectivity was verified.